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Fundamentals of Red Hat OpenShift Administration

Overview

Course

Guided Exercises

Virtual lab environments are not active for this trial. Exercises and their steps are included for informational purposes only. View the video to see a demonstration of each exercise by one of our instructors.

Close

Guided Exercise: Navigate the OpenShift Web Console

Access an OpenShift cluster by using its web console and review pages to identify key OpenShift cluster services.

Outcomes

- Explore the features and components of Red Hat OpenShift by using the web console.
- Create a sample application by using the Developer perspective in the web console.
- Switch to the Administrator perspective and examine the resources that are created for the sample application.
- Use the web console to describe the cluster nodes, networking, storage, and authentication.
- View the default cluster operators, pods, deployments, and services.



As the student user on the workstation machine, use the `lab` command to prepare your system for this exercise.

This command ensures that the cluster is validated for the exercise.

```
[student@workstation ~]$ lab start intro-navigate
```

Instructions

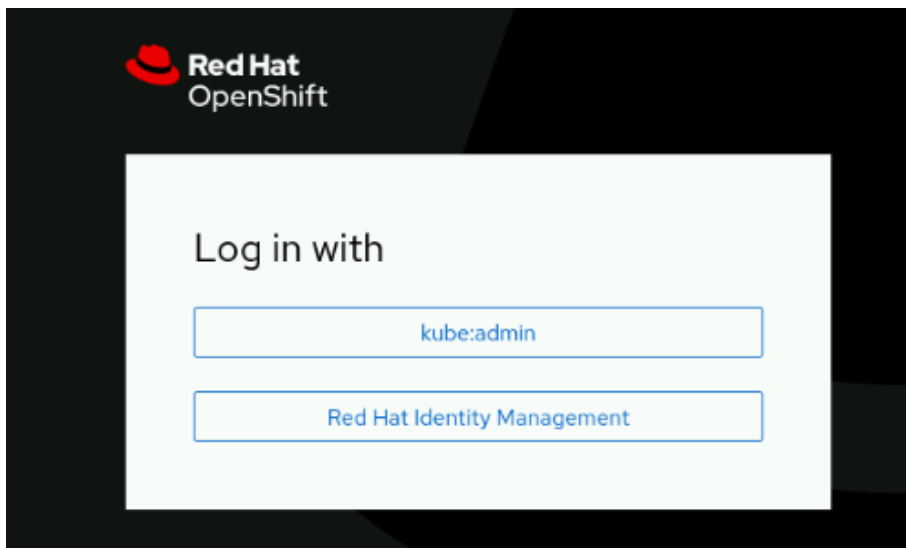
1. As the developer user, locate and then navigate to the Red Hat OpenShift web console.
- 1.1. Use the terminal to log in to the OpenShift cluster as the developer user with the developer password.

```
[student@workstation ~]$ oc login -u developer -p developer \
https://api.ocp4.example.com:6443
...output omitted...
```

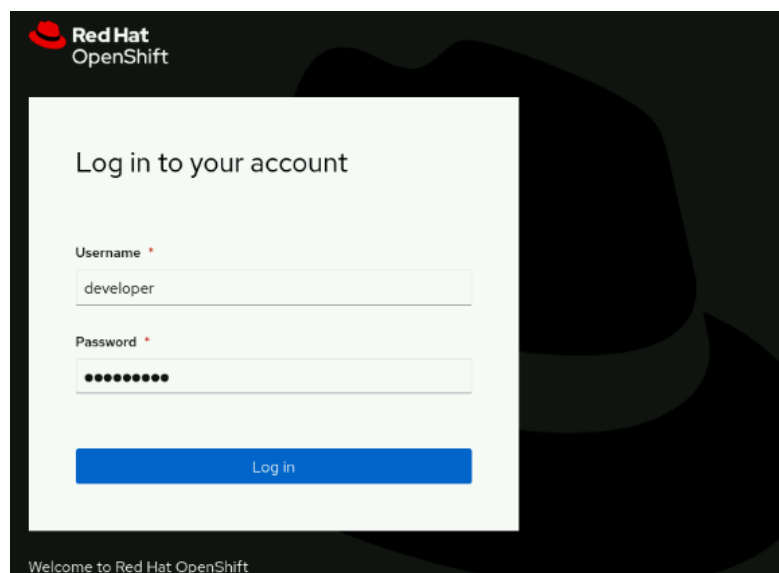
- 1.2. Identify the URL for the OpenShift web console.

```
[student@workstation ~]$ oc whoami --show-console
https://console-openshift-console.apps.ocp4.example.com
```

- 1.3. Open a web browser and navigate to <https://console-openshift-console.apps.ocp4.example.com>. Either type the URL in a web browser, or right-click and select `Open Link` from the terminal.

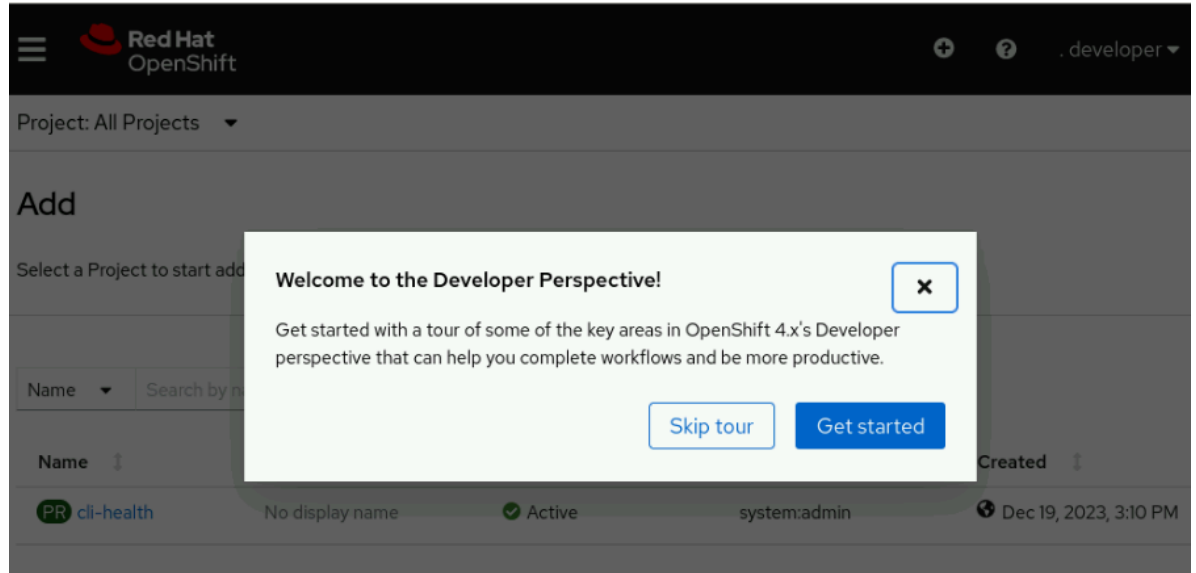


2. Log in to the OpenShift web console as the developer user.
- 2.1. Click **Red Hat Identity Management** and log in as the developer user with the developer password.



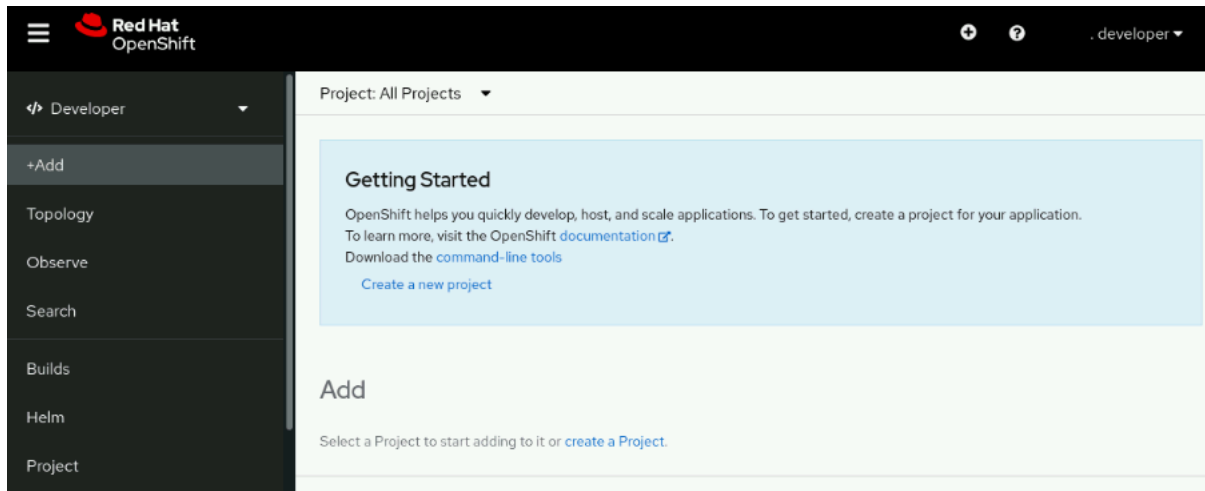
NOTE

Click **Skip Tour** to dismiss the option to view a short tour on the first visit.



3. Use the Developer perspective of the web console to create your first project.

3.1. From the Getting Started section in the +Add page, click **Create a new project** to open the Create Project wizard.



3.2. Create a project named intro-navigate by using the wizard. Use intro-navigate for the display name, and add a brief description of the project.

Create Project

An OpenShift project is an alternative representation of a Kubernetes namespace.

[Learn more about working with projects](#)

Name * ⓘ

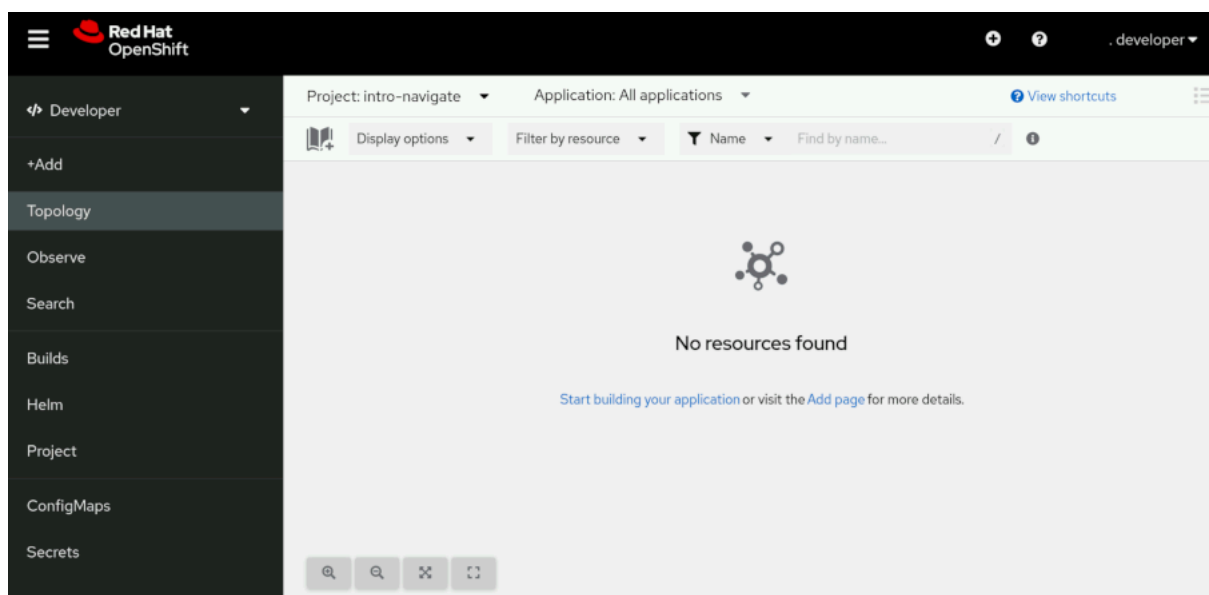
Display name

Description

[Cancel](#) [Create](#)

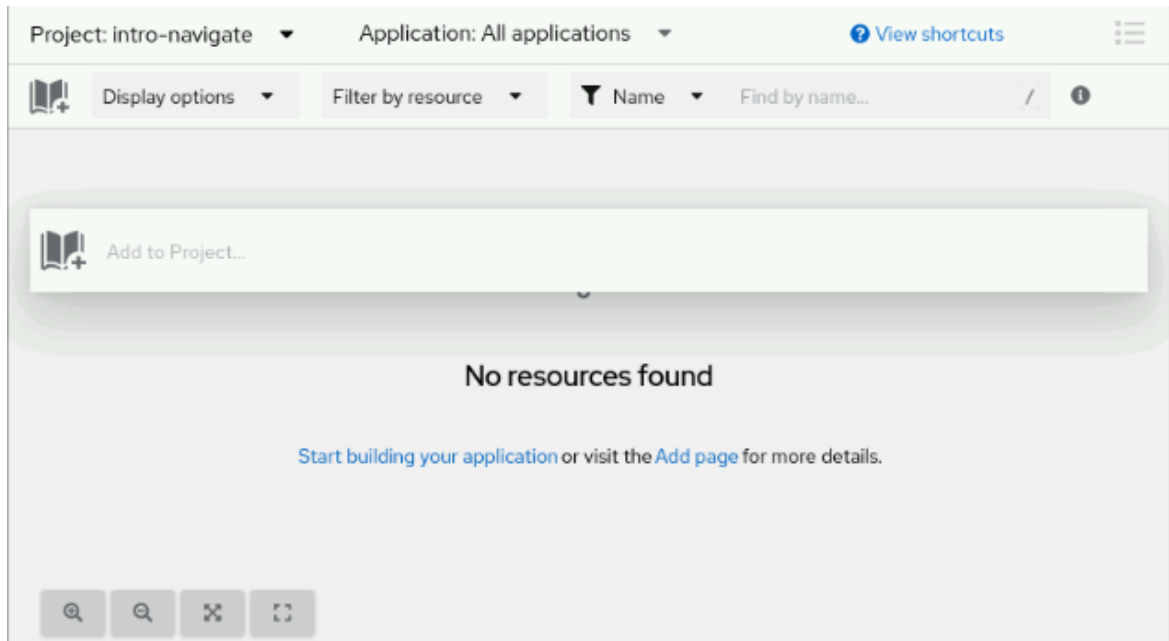
3.3. Click **Create** to create the project.

3.4. Click **Topology** to view the project.

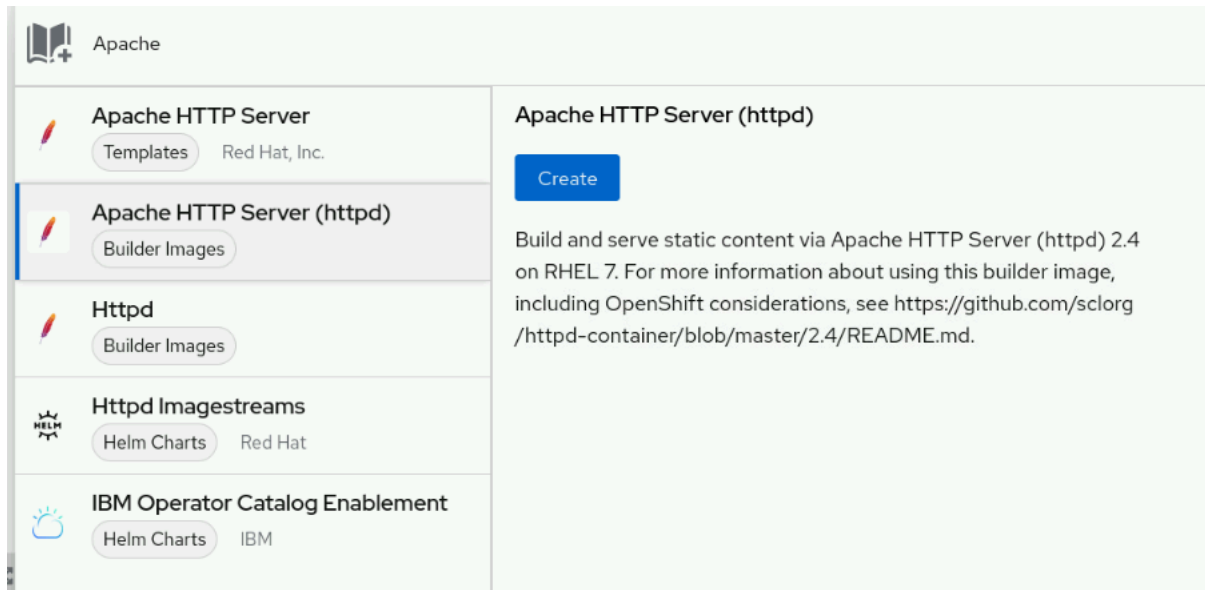


4. Deploy a sample application in the project.

4.1. Select the **Start building your application** link to browse the available sample applications.



4.2. Enter Apache into the search bar to see the available sample applications for deployment.



- 4.3. Select the samples option from **Apache HTTP Server (httpd)** from the list of available applications, and then click **Create** from the side panel.
- 4.4. Use application-test for the application name and the name of the deployment, `https://git.ocp4.example.com/developer/httpd-ex.git` for the git repository URL, and GitLab for the git type. Leave the rest of the default values, and select **Create** at the bottom of the page to create the application.

Project: intro-navigate Application: All applications

Apache HTTP Server 2.4 (RHEL 7)

BUILDER HTTPD

Build and serve static content via Apache HTTP Server (httpd) 2.4 on RHEL 7. For more information about using this builder image, including OpenShift considerations, see <https://github.com/sclorg/httpd-container/blob/master/2.4/README.md>. Sample repository: <https://github.com/sclorg/httpd-ex.git>

Git

Git Repo URL *

`https://git.ocp4.example.com/developer/httpd-ex.git` Validated

Try sample ↗

Git type *

GitLab

Show advanced Git options

General

Application name

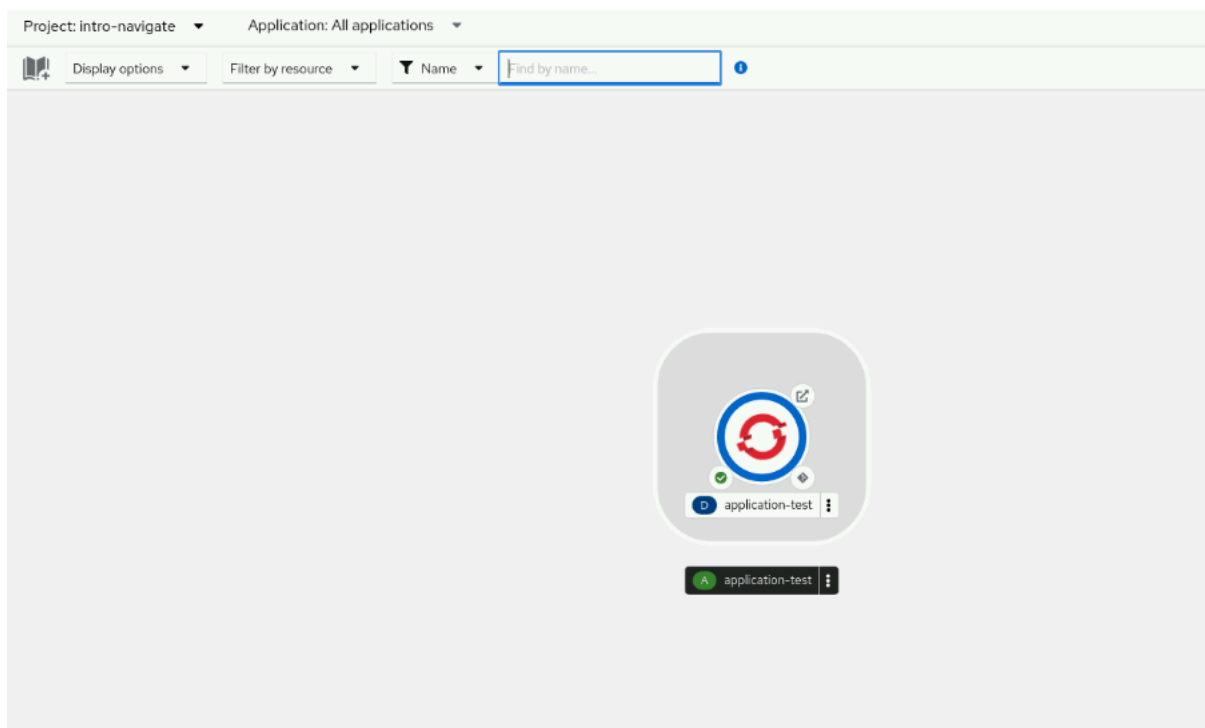
application-test

A unique name given to the application grouping to label your resources.

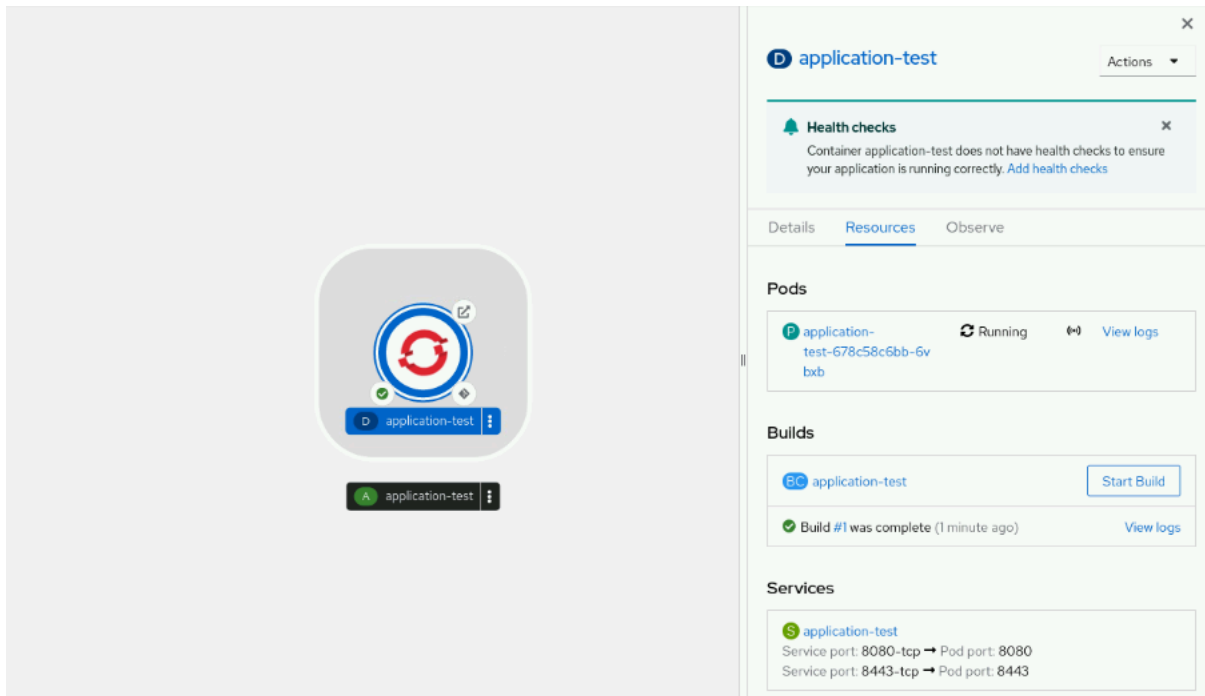
Name *

Create Cancel

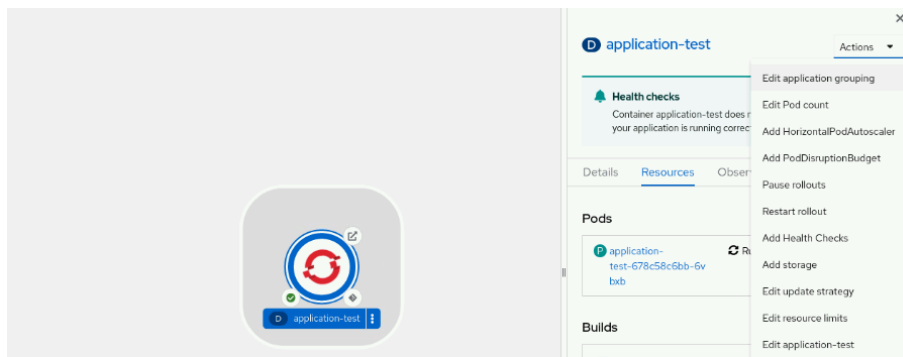
The Topology page opens and displays the application-test application deployment.



5. View the deployment details for the application-test application.
 - 5.1. Select the application-test deployment to view the details of the application.

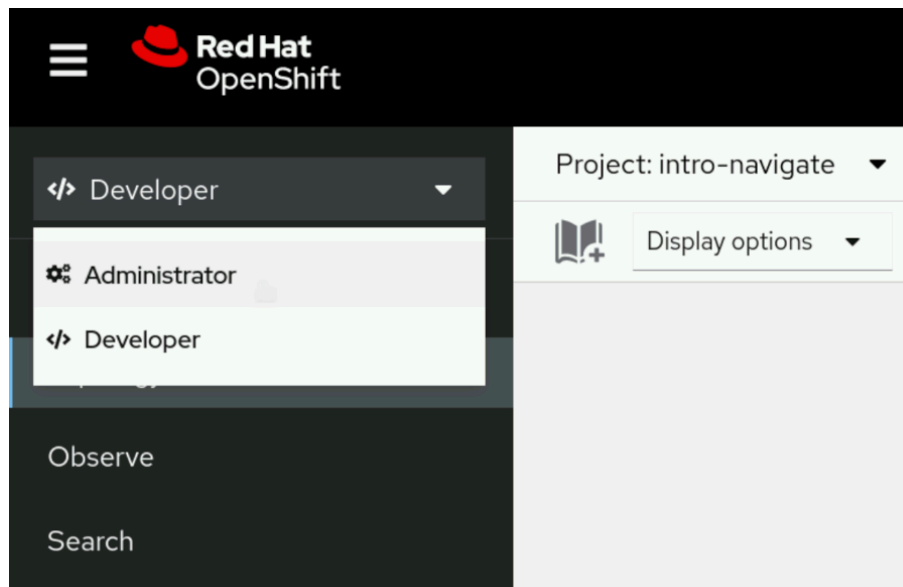


5.2. Select the **Actions** list to view the available controls for the application-test deployment.

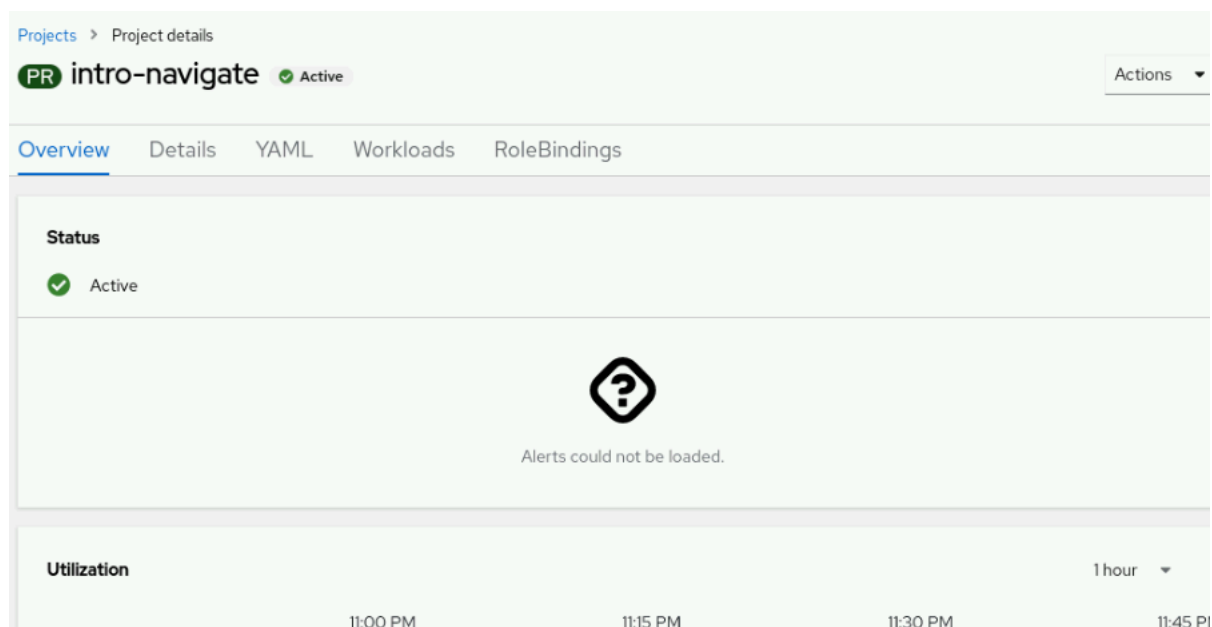


6. Switch into the Administrator perspective and inspect the deployment.

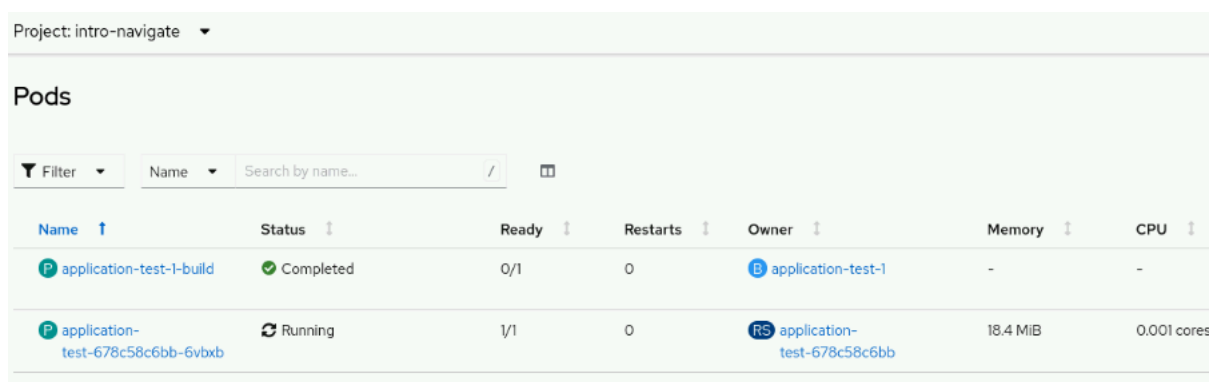
6.1. From the OpenShift web console, locate the left panel. If you do not see the left panel, then click the main menu icon at the upper left of the web console. Click **Developer** and then click **Administrator** to change to the Administrator perspective. The web console changes to the new perspective and exposes additional information through the sidebar.



- 6.2. Select the **intro-navigate** project to open the Project Details page. This page includes a general overview of the project, such as the project status and resource usage details.



7. View the application-test pods and deployment.
 - 7.1. From the OpenShift web console menu, navigate to **Workloads** → **Pods** to view the application-test pods.



- 7.2. Navigate to **Workloads** → **Deployments** to view the list of deployments in the project.

Project: intro-navigate

Deployments

Name Search by name... /

Name	Status	Labels	Pod selector
 application-test	1 of 1 pods	app=application-test app.kubernetes.io/component=application-test app.kubernetes.io/instance=application-test app.kubernetes.io/name=application-test app.kubernetes.io/part-of=application-test app.openshift.io/runtime=httpd app.openshift.io/runtime-version=2.4-el7	 app=application-test

7.3. Click **application-test** to view the deployment details.

Project: intro-navigate

Deployments > Deployment details

application-test

Details Metrics YAML ReplicaSets Pods Environment Events

Deployment details

 1 Pod

Name application-test	Update strategy RollingUpdate
Namespace  intro-navigate	Max unavailable 25% of 1 pod

8. View the application-test service and route.

8.1. Navigate to **Networking** → **Services** and click **application-test** to view the details of the application-test service.

Project: intro-navigate

Services

Name Search by name... /

Name	Labels	Pod selector	Location
 application-test	app=application-test app.kubernetes.io/component=application-test app.kubernetes.io/instance=application-test app.kubernetes.io/name=application-test app.kubernetes.io/part-of=application-test app.openshift.io/runtime=httpd app.openshift.io/runtime-version=2.4-el7	 app=application-test, deployment=application-test	172.30.215.52:80 172.30.215.52:84

8.2. Navigate to **Networking** → **Route** and click **application-test** to view the details of the application-test route.

Project: intro-navigate ▾

Routes

Filter ▾ Name ▾ Search by name... /

Name ▴	Status ▴	Location ▴	Service ▴
 application-test	 Accepted	https://application-test-intro-navigate.apps.ocp4.example.com  	 application-test

9. Delete the project and log out of the web console.

9.1. From the OpenShift web console menu, navigate to **Home** → **Projects**. Select **Delete Project** from the context menu for the intro-navigate project.

Administrator ▾
Home ▾
Projects
Search
API Explorer
Events
Operators >

Projects

Create Project

Filter ▾ Name ▾ Search by name... /

Name ▴	Display name ▴	Status ▴	Requester ▴	Created ▴
 intro-navigate	No display name	 Active	developer	Jan 3, 2024, 9:39 PM

Edit Project
Delete Project

9.2. Enter the project name in the text field and then select **Delete**.



Delete Project?

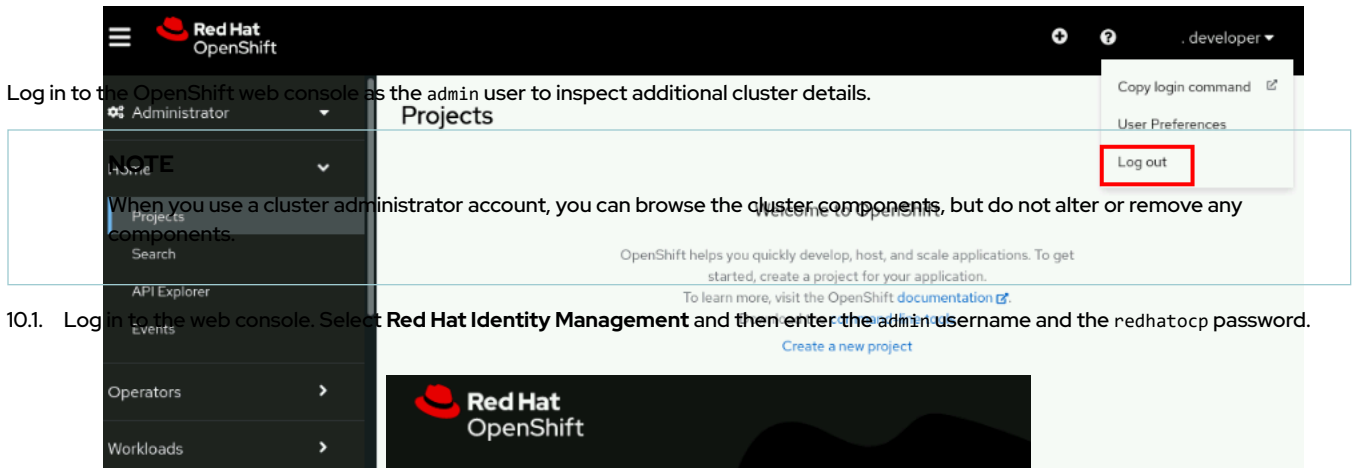
This action cannot be undone. It will destroy all pods, services and other objects in the namespace **intro-navigate**.

Confirm deletion by typing **intro-navigate** below:

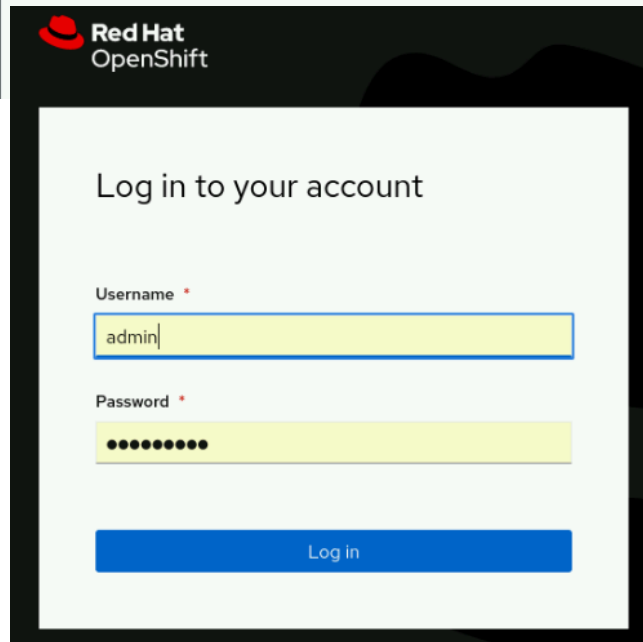
Cancel
Delete

9.3. Log out of the web console. From the OpenShift web console right panel, click **developer** and then select **Log out** from the account menu.

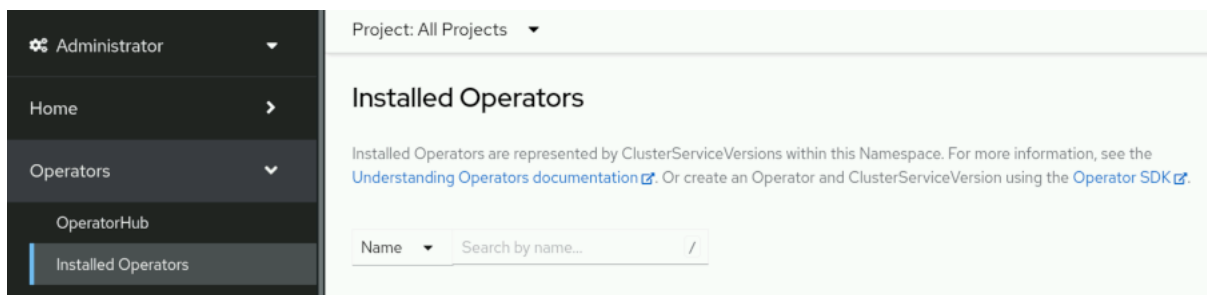
10. Log in to the OpenShift web console as the admin user to inspect additional cluster details.



- 10.1. Log in to the web console. Select **Red Hat Identity Management** and then enter the admin username and the redhatocp password.



- 10.2. From the OpenShift web console menu, navigate to **Operators** → **Installed Operators**. Each operator provides a specific function for the cluster. Select an individual operator to display its details.



- 10.3. Navigate to **Workloads** → **Pods** to view the list of all pods in the cluster. The search bar at the top can narrow down the list of pods. Select an individual pod to display its details.



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